ETHNOHACK 2026 · TRACK 1 · TRADITIONAL KNOWLEDGE

Plant librarian

Traditional plant knowledge from Kosovo and Albania, read from the studies, checked against modern science, and every line linked to its source.

Rizonë Hyseni · Dua Bilalli · Sara Kycyku · Rrezon Ibishi

plant-librarian.streamlit.app

THE PROBLEM

The knowledge exists. Nobody connects it.

- **It lives with elderly people.** The Sharr Mountains study interviewed 139 elderly informants from three communities.
- **It is scattered** across studies with different tables and formats.
- **Names change by village.** *Hithi*, *Kopriva* and *Yakici* are the same plant: nettle.
- **Science tests something else.** People drink a tea; papers test essential oils and alcohol extracts.

“Was the tea my grandmother drinks ever tested?”

Answering this today takes a student days: find the studies, decode the dialect name, search PubMed plant by plant, and check every paper's preparation.

WHAT WE BUILT

From a study PDF to a sourced fact sheet

1 · Read
An AI reads open-access studies into rows: plant, part, preparation, use, page and exact quote.

→

2 · Ground
A row is kept only if its quote is found on the cited page of the PDF.

→

3 · Name
Dialect names matched with fuzzy search, every plant confirmed by GBIF.

→

4 · Compare
PubMed, GBIF, Wikidata compounds, EMA safety rules. Was the same preparation tested?

→

5 · Show
A bilingual app: profiles, map, gaps, priorities, chat.

AI **Only where text must become a table**

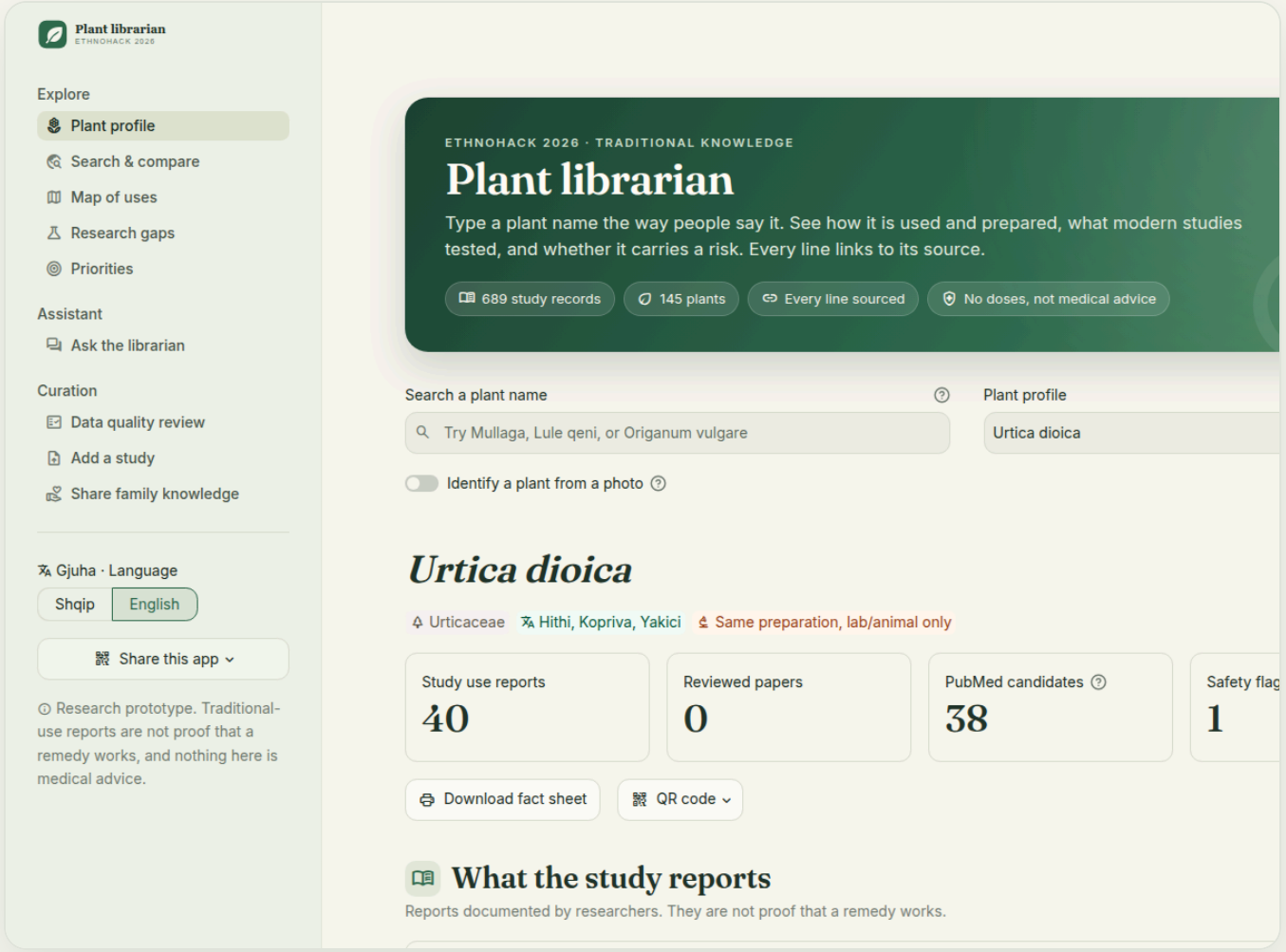
Reading studies, transcribing voice, answering questions from our own fact sheets.

Rules **Everything that must be right**

Safety flags, evidence verdicts, priorities and knowledge-at-risk come from written rules and open databases, never from an AI guess.

LIVE APP · PLANT PROFILE

Type the name the way people say it



Hithi* → *Urtica dioica (nettle), used by Albanian, Bosniak/Gorani and Turkish families.

- Every use shows the **page and exact quote** from the study.
- **Was the same preparation tested?** 38 PubMed candidates; the tea only in lab or animals.
- Safety flags from EMA rules. **Never a dose.**
- Identify a plant from a **photo** (Pl@ntNet), print a **fact sheet**, share by **QR**.
- Albanian and English.

THE DATA

Two open studies from Kosovo, one library

689
use records, each with page and quote

145
plants, all names confirmed by GBIF

3
communities: Albanian, Bosniak/Gorani, Turkish

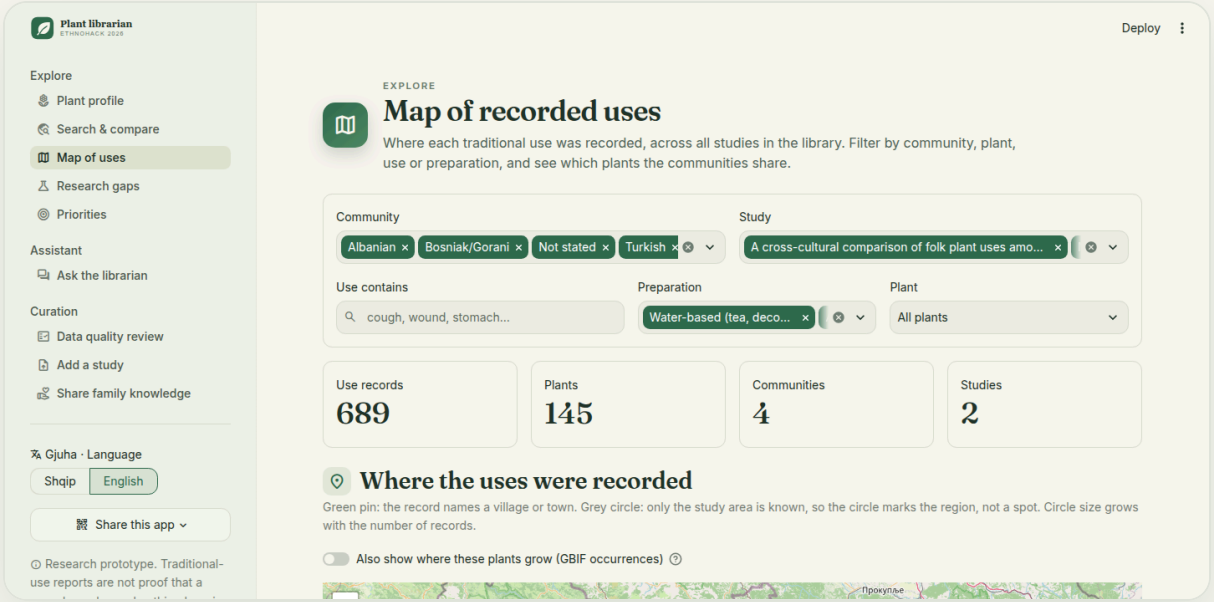
141
plants with PubMed candidates

Sharr Mountains · Mustafa et al. 2015

569 records, 108 plants. Parsed from the study's table without AI. CC BY 4.0.

Albanian Alps, Rugovë · Mustafa et al. 2012

120 records, 88 plants. Extracted with the free Groq LLM. CC BY 2.0.



FINDING 1 · WAS THE SAME PREPARATION TESTED?

17 plants are research gaps

Same preparation, tagged as a human study	23
Same preparation, lab or animal only	71
Only other preparations tested	14
No studies found	3
Manually reviewed	3
Not comparable by keywords	31

Keyword screening of PubMed candidates: a pointer to what to read next, not a systematic review.

Bari sherbetit · hop (*Humulus lupulus*)

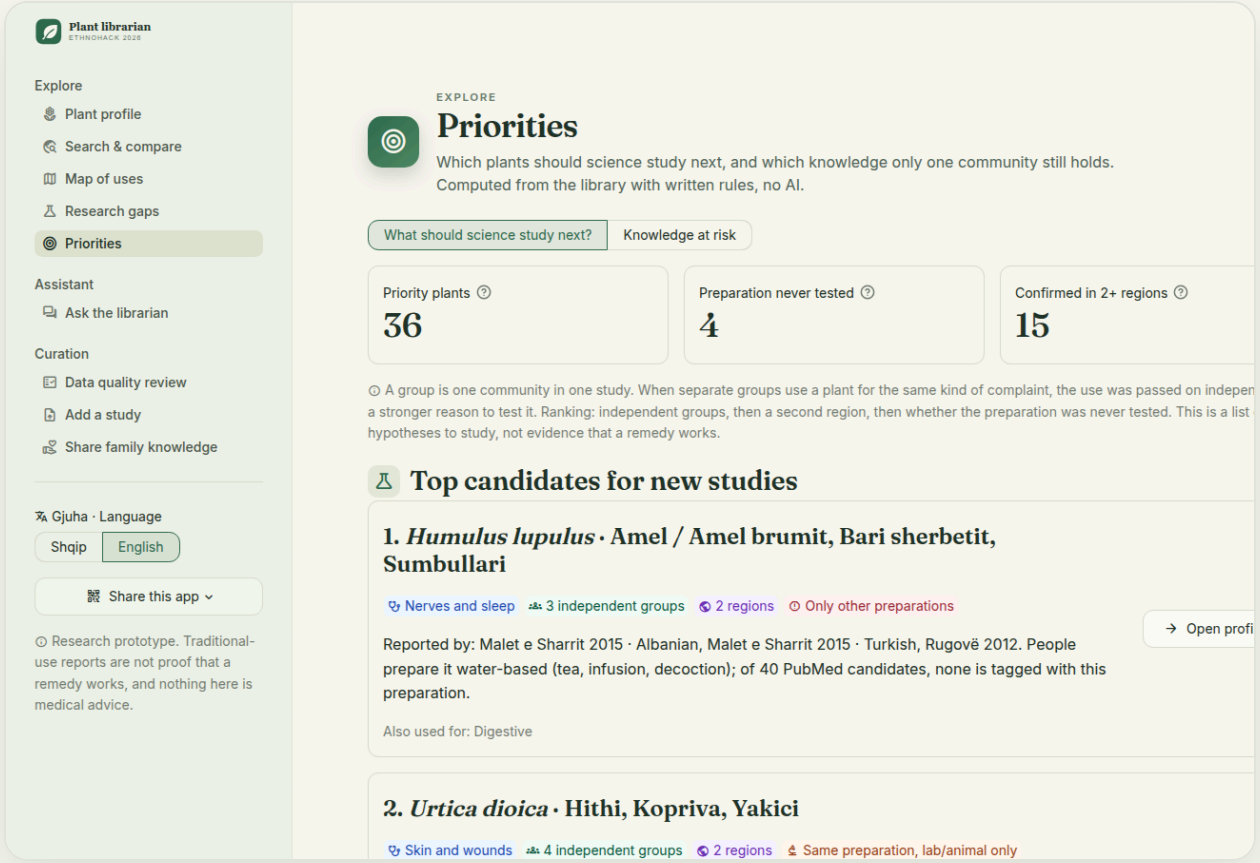
Albanian and Turkish families in Sharr and Albanians in Rugovë drink it as a **tea** to relax and sleep.

Of **40** PubMed papers, **none** tests the tea: essential oils, alcohol extracts, isolated compounds.

A concrete question a pharmacy student can take on.

FINDING 2 · WHAT TO STUDY NEXT, AND WHAT COULD BE LOST

36 plants to test next · 73 at risk of being lost



Research priorities

Plants used for the same complaint by **2+ independent groups**, whose preparation was never tested in people. 15 confirmed in both regions. #1: hop, for nerves and sleep.

Knowledge at risk

73 plants and **119 uses** are reported by one group only. If that group stops passing them on, nothing else records them.

Shared heritage

18 plants are used by all three Sharr communities: nettle, chamomile, elder, linden, juniper…

WHY YOU CAN TRUST IT

Every line sourced, every number checked

95%
recall on 42 hand-checked rows (40/42)

677
of 689 records pass the automatic PDF check

21
plants with an EMA/HMPC safety flag

0
doses shown, anywhere

Grounded

An extracted row is dropped if its quote is not on the cited page. The 12 failing records are flagged for a human to fix.

Human in the loop

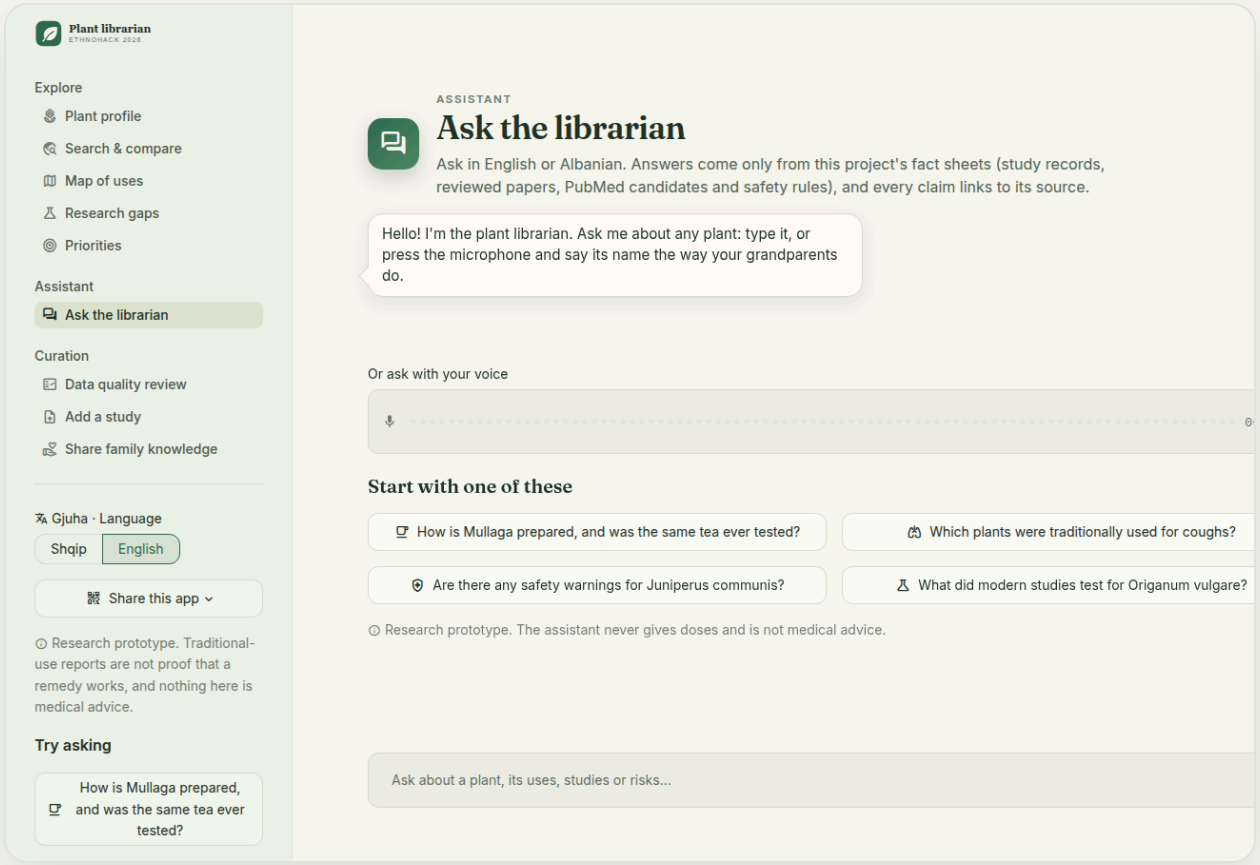
Records stay “Needs review” until a named person verifies them. A machine check never counts as a human one.

Reproducible

65 tests and CI. API responses are cached: one command rebuilds the whole library offline.

FOR EVERYONE, NOT ONLY RESEARCHERS

Ask, speak, photograph, share



- **Ask the librarian** in Albanian or English. Answers only from our fact sheets, each claim cited, no doses.
- **Voice questions**: say the plant's name the way your grandparents do.
- **Photo identification** with Pl@ntNet, shown as a suggestion with a score.
- **Share family knowledge**: two consents, no names, doses refused, published only after review and withdrawable.
- **Add a study**: upload a new PDF and it appears on the map.

Built only on free services: Groq, GBIF, PubMed, Wikidata, Pl@ntNet, Streamlit Cloud.

WHAT'S NEXT

A map of what to test next

- More open studies from Kosovo and Albania (Gollobordo, southern Kosovo).
- Verify records with local experts and universities.
- Conservation status: is the plant itself at risk?
- Give the priority list to pharmacy and biology students as thesis topics.
- Collect family knowledge from schools and villages, with consent.

Traditional use is not proof that a remedy works. This is a research prototype, not medical advice.

Try it: plant-librarian.streamlit.app

github.com/rizonehyseni/Plant-Librarian

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